

James A. Nguyen

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EDUCATION

University of California, Irvine, CA
B.S., Applied and Computational Mathematics
Concentration in Data Science, Minor in Statistics

Sep 2022 - Sep 2025
3.76 GPA

Relevant Coursework: Numerical Linear Algebra, Deep Learning and Neural Networks, Stochastic Processes, Real and Complex Analysis, Mathematical Optimization

RESEARCH EXPERIENCE

Undergraduate Research Assistant, Statistics
Donald Bren School of Information and Computer Sciences, UC Irvine

Jun 2025 - Present

- Among 16 undergraduates admitted to the Summer Institute for Biostatistics (SIBS, NIH/NIAID REU)
- Created pipeline in RStudio to process patient data and analyze effect of HIV on Tuberculosis treatment outcomes before and after universal access to Antiretroviral Therapy in Botswana
- Produced pointwise metrics to evaluate spatial correlation between individual participants and justify assumptions of independence in regression modeling
- Tuned random forest to identify nonlinear effects by variable importance plots

Undergraduate Researcher, Mathematics
School of Physical Sciences, UC Irvine

Sep 2024 - Sep 2025

- Proposed and optimized alternative sampling approaches using MATLAB to row-iterative projection methods for solving large-scale linear feasibility problems
- Developed algorithm to cluster within linear systems to diversify sampling by orthogonality
- New methods promote 100 percent faster convergence and better approximation error than Sampling Kaczmarz-Motzkin algorithm applied to the common case of ill-conditioned and overdetermined systems

Research Assistant, Data Science
Fortin Lab - Center for the Neurobiology of Learning and Memory

Jun 2024 - Jan 2025
Irvine, CA

- Among only 6 undergraduates selected for funded position to continue research beyond SoCal Data Science Research Fellowship (NSF REU)
- Developed and applied pointwise sliding-window analysis of correlation in Python for multinomial sequence detection in neuronal firing ensembles
- Analyzed KL-divergence between Poisson-simulated hypothesis distribution and observed decoding probability distribution

PUBLICATIONS, PRESENTATIONS

A. Baniasad*, S. Chao*, J. Nguyen*, E. Tian*, C. Modongo, V. Minin, J. Sebastian, and S. Shin. HIV Remains a Risk Factor for Unfavorable Tuberculosis Treatment Outcomes in the Era of Universal Access to Antiretroviral Therapy in Botswana, medRxiv, 340699, 2025, preprint available at doi.org/10.1101/2025.11.26.25340699.

J. Nguyen*, O. Presnyakov*, A. Radhakrishnan*. Read Between the Hyperplanes: On Spectral Projection and Sampling Approaches to Randomized Kaczmarz, arXiv, 2025, preprint available at doi.org/10.48550/arXiv.2511.03055.

* - joint first authors

TECHNICAL SKILLS

Git, Google Cloud Platform, LaTeX, MATLAB, NumPy, Pandas, Plotly, Python, PyTorch, R/RStudio, Scikit-Learn, SQL, TensorFlow, Tidyverse, VS Code, Matplotlib